

MedTrack DV — Module 7

Dashboard Testing Report | Testing and Validation

Project: MedTrack_DV (Hospital Operations & Patient Analytics Dashboard)

Final workbook under test: MedTrack_DV.twbx

1. Testing Objective

The objective of Module 7 testing is to validate KPI calculations, verify healthcare operational metrics, test dashboard interactions, and validate patient-flow analytics in the final integrated Tableau dashboard suite.

2. Testing Scope

Area	Scope covered
KPI validation	Seven implemented KPIs: Total Admissions, Occupancy Rate, Average Length of Stay, Readmission Rate, Bed Utilization Rate, Department Efficiency Score, and Staff Utilization Rate.
Healthcare metrics	Admissions, length of stay, readmission, bed occupancy/utilization, staff utilization, and department efficiency.
Dashboard testing	Hospital Overview, Patient Flow, Department Analytics, and Resource Utilization.
Patient-flow validation	Admission analysis, monthly patient load, discharge trend, transfers by department, and average length of stay.
Interactions	State and Department filters, cross-filtering, department selection parameter action, and dashboard navigation/integration.

3. KPI Calculation Validation

The KPI-generation implementation was reviewed against the final dataset. The following values were reproduced from the final dataset using the implemented definitions.

KPI	Validation basis	Validated value	Status
Total Admissions	Record count	100,000	PASS
Occupancy Rate	Mean of Bed_Occupancy_Rate_Calc × 100	36.757%	PASS
Average Length of Stay	Mean of Length of Stay	7.50314 days	PASS
Readmission Rate	Readmitted patients / total admissions × 100	7.477%	PASS
Bed Utilization Rate	Occupied beds / total admissions × 100	50.36%	PASS
Department Efficiency Score	Mean of implemented efficiency component	49.35023%	PASS
Staff Utilization Rate	Mean of Staff_Utilization_Calc	57.92486%	PASS

KPI validation result: 7/7 implemented KPI calculations were reproduced successfully from the final dataset. This is 100% formula-level/numerical reproducibility for the implemented KPI definitions.

Accuracy qualification: The project requirement specifies KPI accuracy above 95%. The 7/7 result demonstrates reproducibility of the implemented calculations; it is not an independent statistical accuracy measurement against an external ground-truth dataset. No unsupported accuracy percentage has been claimed.

4. Healthcare Metric Verification

Metric	Verification performed	Result
Admissions	Checked total admissions KPI and admission-related dashboard views.	PASS
Length of Stay	Checked LOS calculation in days and its use in dashboard analysis.	PASS
Readmission	Checked overall readmission KPI and department-level readmission analysis.	PASS
Bed occupancy/utilization	Checked occupancy/utilization KPIs and resource dashboard views.	PASS

Metric	Verification performed	Result
Staff utilization	Checked implemented staff-utilization KPI and department-level resource view.	PASS
Department efficiency	Checked Department Efficiency Score and department comparison view.	PASS

5. Dashboard Functional Testing

Dashboard	Validation performed	Result
Hospital Overview	Verified KPI cards, department patient volume, hospital-type distribution, state volume, and monthly patient load analysis.	PASS
Patient Flow	Verified admission analysis, transfers by department, monthly discharge trend, patient load, and LOS analysis.	PASS
Department Analytics	Verified department patient volume, readmission, LOS, efficiency score, and treatment capacity analysis.	PASS
Resource Utilization	Verified staff utilization, equipment utilization, resource availability, capacity planning, and bed occupancy analysis.	PASS

6. Patient Flow Analytics Validation

Test	Expected / required analysis	Result
Admission analysis	Admission-related patient analysis is available.	PASS
Patient movement	Transfers by department are available.	PASS
Discharge tracking	Monthly discharge trend is available.	PASS
Average stay	Average length of stay is available for analysis.	PASS
Patient load	Monthly patient load trend is available.	PASS

7. Dashboard Interaction Testing

Interaction	Expected behavior	Result
State filter	Filters the dashboard analysis by selected state.	PASS
Department filter	Filters the dashboard analysis by selected department.	PASS
Cross-filtering	Chart selections affect related views where configured.	PASS
Department parameter action	Department selection is reflected in the selected-department analysis/display.	PASS
Navigation	Navigation controls switch among the four dashboards.	PASS
Dashboard integration	The four dashboards operate as one integrated workbook.	PASS

8. Overall Test Result

Requirement	Result
KPI calculations validated	PASS — 7/7 implemented KPI calculations reproduced.
Healthcare metrics verified	PASS — operational metrics verified against the implemented dataset and dashboard usage.
Dashboard interactions tested	PASS — filters, cross-filtering, parameter action, navigation and integration covered.
Patient flow analytics validated	PASS — admissions, transfers, discharges, LOS and patient-load analysis covered.
Major dashboard issues	No major issues identified within the documented validation scope.

9. Conclusion

The final MedTrack DV workbook satisfies the Module 7 testing scope. KPI calculations were reproduced successfully, healthcare operational metrics were verified, patient-flow analytics were checked, and the implemented dashboard

interactions were validated. The documented QA result is PASS with no major dashboard issues identified.

The KPI requirement of greater than 95% is addressed through reproducibility validation of the implemented calculations. Because no independent external ground-truth dataset was supplied, the report does not claim a fabricated statistical accuracy percentage.